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Low-Speed High-Torque Drive Sizing Worksheet

For OEM designers and maintenance engineers sizing geared PMDC or BLDC drives for slow, heavily loaded shafts

Work through the steps in order and record each value in the blanks. Use the loaded motor speed, not the no-load speed, and confirm final figures against the motor and gearbox data sheets before ordering.

Step 1 — Define the output requirement

- [ ] Continuous output torque at the operating point: _____ Nm
- [ ] Required output speed under load: _____ rpm, tolerance ± _____ %
- [ ] Peak or breakaway torque: _____ Nm for _____ s, _____ times per hour
- [ ] Duty type to IEC 60034-1: S1 continuous / S2 short-time (minutes) / S3 intermittent (% on-time)
- [ ] Holding requirement: none / hold on power loss / suspended or inclined load
- [ ] Output connection: pulley or sprocket pitch diameter _____ mm, distance from gearbox face _____ mm
- [ ] Maximum ambient temperature at the motor: _____ °C (allow for 45–50 °C summer conditions)

Step 2 — Calculate output power and choose motor base speed

- [ ] Output power P = 2 × 3.1416 × N × T / 60 (W), with N in rpm and T in Nm
- [ ] Reference check: 50 Nm at 30 rpm = 157 W
- [ ] Choose a loaded motor speed, typically 1,500–3,000 rpm for brushed PMDC motors
- [ ] Do not reach the target speed by voltage or PWM reduction: gear any permanent reduction of more than about 20–30 % of rated speed
- [ ] Higher motor speed gives a smaller motor but a larger ratio, more stages and lower gearbox efficiency

Step 3 — Calculate ratio and gearbox efficiency

- [ ] Ratio i = loaded motor speed / required output speed = _____
- [ ] Round i to a standard ratio and recalculate output speed; confirm it is within tolerance
- [ ] Gearbox efficiency = (efficiency per stage) raised to the number of stages, e.g. 0.95 × 0.95 × 0.95 = 0.857
- [ ] For worm gearing use the maker's figure for the actual ratio; high ratios are often 40–60 %
- [ ] Gearbox heat (W) = motor shaft power - output power; check it against the gearbox thermal rating on S1 duty

Step 4 — Check the motor

- [ ] Required continuous motor torque Tm = output torque / (i × gearbox efficiency) = _____ Nm
- [ ] Motor rated continuous torque at the loaded speed should be at least 1.2 × Tm
- [ ] Peak motor torque for breakaway = peak output torque / (i × gearbox efficiency); confirm the controller current limit allows it
- [ ] Motor shaft power = Tm × motor rpm / 9.55 (W); electrical input = shaft power / motor efficiency
- [ ] Plot the operating point on the output torque–speed curve: continuous point inside the continuous zone, breakaway inside the intermittent zone

Step 5 — Check stall, brake and shaft loads

- [ ] Stall case: motor stall torque × i × gearbox efficiency must not exceed the gearbox peak rating, or fit a current limit or slip clutch
- [ ] Brake torque at the motor shaft ≥ 2 × output holding torque / i (gear efficiency taken as 1 for safety)
- [ ] Do not rely on worm self-locking for suspended or inclined loads; fit a fail-safe brake
- [ ] Radial shaft load F = 2 × output torque / pitch diameter × transmission factor (about 1.0 for chain, higher for belts; Nm and m give N)
- [ ] Compare F with the gearbox overhung load rating at the actual load position; mount the pulley or sprocket close to the housing

Typical Gear Type Characteristics (per stage)

Gear type	Ratio per stage	Efficiency per stage	Back-driving	Typical use
Spur	Up to about 8:1	95–98 %	Freely back-drivable	Cost-sensitive parallel-shaft drives
Helical	Up to about 8:1	95–98 %	Freely back-drivable	Quiet continuous and traction drives
Planetary	3:1 to 10:1	About 95–97 %	Freely back-drivable	High torque density, in-line output
Worm	5:1 to 60:1 or more	Roughly 40–90 %	Reduced or locked at high ratios	Right-angle, intermittent duty

Values are typical engineering ranges for planning. Confirm ratings, efficiencies and overhung load limits against the specific motor and gearbox data sheets, or send the completed worksheet to Globe Scott Motors for a sizing review.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.